1 Parametric Equations and Conic Sections

Central question. How can the same curve be described by coordinates, by a moving point, or by a geometric rule?

1.1 A moving point traces a curve

In a parametric description, both coordinates depend on a parameter t:

$$\boxed{x = f(t), \qquad y = g(t).}$$

Think of t as time when a particle moves. The curve records its location, while the parameterization also records direction and timing.

Example 1.1. For $x = 2t + 1$, $y = t - 3$, eliminating $t = (x - 1)/2$ gives $y = (x - 7)/2$. If t increases, the point moves up and right along this line.

1.2 Eliminating the parameter

For $x = 3\cos t$, $y = 2\sin t$, square the normalized equations and add:

$$\boxed{\frac{x^2}{9} + \frac{y^2}{4} = 1.}$$

The equation gives the whole ellipse when t ranges over $[0, 2\pi]$. A restricted interval, such as $[0, \pi]$, traces only its upper half. Different parameterizations may have the same image but different speeds or directions.

Example 1.2. Both $(x, y) = (\cos t, \sin t)$ and $(x, y) = (\cos 2t, \sin 2t)$ trace the unit circle for $0 \le t \le 2\pi$; the second makes two revolutions.

1.3 Projectile motion as a parametric curve

Under constant downward acceleration g, a projectile launched from (x_0, y_0) at speed v_0 and angle θ satisfies

$$x(t) = x_0 + v_0 \cos\theta\, t, \qquad y(t) = y_0 + v_0 \sin\theta\, t - \tfrac{1}{2} g t^2.$$

When the horizontal velocity is nonzero, eliminating t gives a parabola. The motion model adds information the Cartesian curve alone does not supply, including direction and location at a given time.

1.4 Conics: circles and ellipses

A *circle* is the set of points at a fixed distance r from a center (h, k):

$$(x - h)^2 + (y - k)^2 = r^2.$$

An *ellipse* can be defined by a constant sum of distances to two foci. In standard form, for $a \ge b > 0$,

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1.$$

Its horizontal semiaxis is a and vertical semiaxis is b; when the major axis is horizontal, focal distance obeys $c^2 = a^2 - b^2$.

Example 1.3. For $x^2/25 + y^2/9 = 1$, the vertices are $(\pm 5, 0)$, and the foci are $(\pm 4, 0)$.

1.5 Parabolas and their focus

A *parabola* is the set of points equidistant from a fixed point (focus) and a fixed line (directrix). The equation

$$(x-h)^2 = 4p(y-k)$$

has vertex (h,k), focus $(h,k+p)$, and directrix $y=k-p$. Its axis is vertical; $p>0$ opens upward and $p<0$ downward.

Example 1.4. For $x^2=8y$, we have $4p=8$, so the focus is $(0,2)$ and the directrix is $y=-2$.

1.6 Hyperbolas and differences of distances

A *hyperbola* consists of points for which the absolute difference of distances to two foci is fixed. A horizontal hyperbola has standard form

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1.$$

Its vertices are $(h\pm a,k)$, its foci are $(h\pm c,k)$ where $c^2=a^2+b^2$, and its asymptotes are $y-k=\pm(b/a)(x-h)$.

Example 1.5. The hyperbola $x^2/9 - y^2/16 = 1$ has vertices $(\pm 3,0)$, foci $(\pm 5,0)$, and asymptotes $y=\pm 4x/3$.

1.7 Changing descriptions

A parametric curve may be easier to generate than to recognize from a Cartesian equation. A Cartesian equation reveals symmetry or an axis. A focal definition explains why the curve arises in optics, navigation, and geometry. None of these descriptions replaces the others; each answers a different question.

1.8 Exercises

1. Eliminate t from $x=t+2$, $y=3t-1$ and describe the motion as t increases.
2. Find the Cartesian equation traced by $x=4\cos t$, $y=3\sin t$.
3. Explain how $(\cos t,\sin t)$ and $(\cos 3t,\sin 3t)$ differ as parameterizations.
4. A projectile leaves the origin at 20 m/s at 30°, with $g=9.8$ m/s^2. Write its parametric equations.
5. Identify center and radius of $(x-2)^2+(y+1)^2=16$.
6. Find the foci of $x^2/36+y^2/20=1$.
7. Find the focus and directrix of $(x-1)^2=12(y+2)$.
8. Find the vertices and asymptotes of $x^2/4-y^2/9=1$.